

6. Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

a)

CGPA	Aptitude Score	Communication Score	Placement
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
df = pd.read_excel("Dataset.xlsx",sheet_name="Dataset_1",nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.3,random_state=42,stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
```

```

print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(5, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=encoder.classes_,
    yticklabels=encoder.classes_
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Naive Bayes - Placement Prediction")
plt.show()
new_sample = [[8.9, 87, 84]]
prediction = model.predict(new_sample)
print("\nNew Sample Prediction:",encoder.inverse_transform(prediction)[0])

```

OUTPUT:

Dataset:

	CGPA	Aptitude Score	Communication Score	Placement
0	9.2	90	88	Placed
1	8.8	85	82	Placed
2	8.5	80	79	Placed
3	7.2	72	70	Not Placed
4	6.8	68	65	Not Placed
5	9.0	92	90	Placed
6	7.5	75	74	Not Placed
7	8.6	84	81	Placed
8	6.5	60	58	Not Placed
9	8.9	88	86	Placed

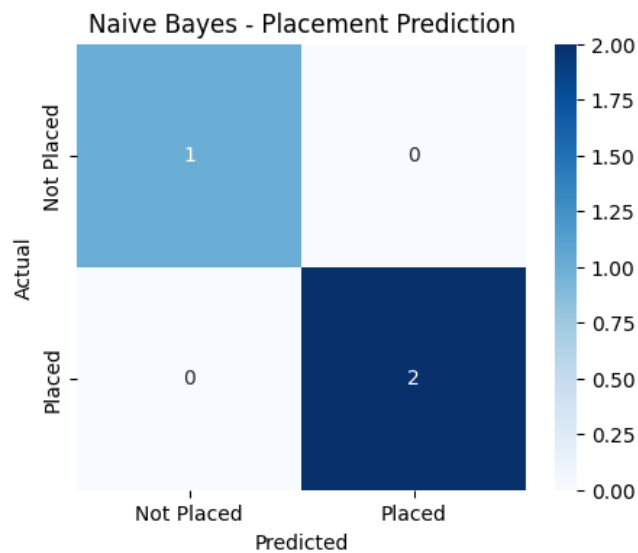
Accuracy: 100.0 %

Confusion Matrix:

```

[[1 0]
 [0 2]]

```



b)

Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
8	780	5	Yes
7	760	4	Yes
6	720	6	Yes
4	620	8	No
3	580	10	No
9	800	4	Yes
5	650	7	No
8	770	5	Yes
4	600	9	No
7	740	6	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
df = pd.read_excel("Dataset.xlsx",sheet_name="Dataset_2",nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
```

```

encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.3,random_state=42,stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(5, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=encoder.classes_,
    yticklabels=encoder.classes_
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Naive Bayes - Loan Approval")
plt.show()
new_sample = [[7, 760, 5]]
prediction = model.predict(new_sample)
print("\nNew Sample Prediction:",encoder.inverse_transform(prediction)[0])

```

OUTPUT:

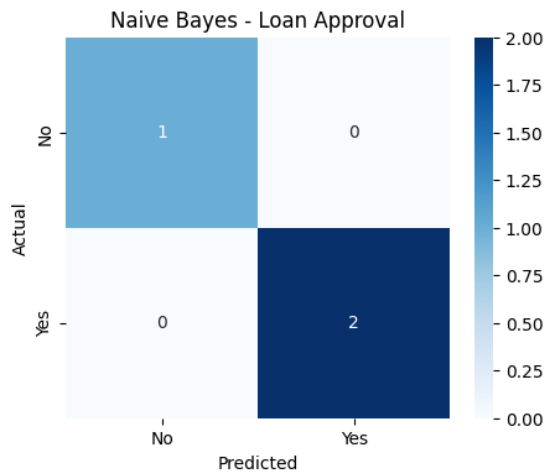
Dataset:

	Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
0	8	780	5	Yes
1	7	760	4	Yes
2	6	720	6	Yes
3	4	620	8	No
4	3	580	10	No
5	9	800	4	Yes
6	5	650	7	No
7	8	770	5	Yes
8	4	600	9	No
9	7	740	6	Yes

Accuracy: 100.0 %

Confusion Matrix:

```
[[1 0]
 [0 2]]
```



c)

Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
39	110	94	Positive
38.5	108	95	Positive
37	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative
38.9	111	93	Positive

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
df = pd.read_excel("Dataset.xlsx",sheet_name="Dataset_3",nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
```

```

X,
y,
test_size=0.3,
random_state=42,
stratify=y
)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(5, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=encoder.classes_,
    yticklabels=encoder.classes_
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Naive Bayes - Disease Prediction")
plt.show()
new_sample = [[38.5, 108, 95]]
prediction = model.predict(new_sample)
print("\nNew Sample Prediction:",encoder.inverse_transform(prediction)[0])

```

OUTPUT:

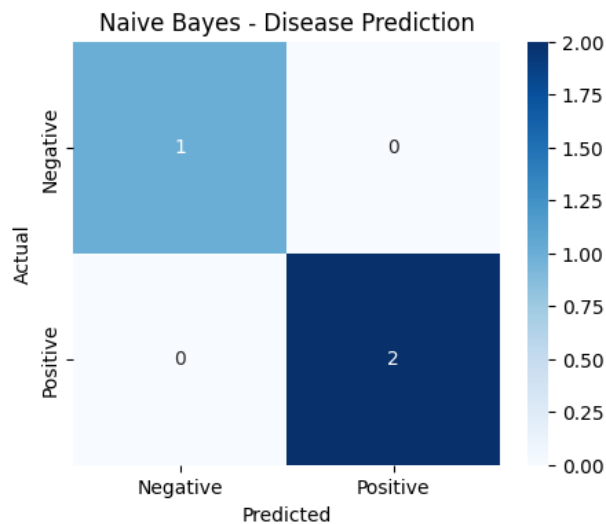
Dataset:

	Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
0	39.0	110	94	Positive
1	38.5	108	95	Positive
2	37.0	78	99	Negative
3	39.2	115	93	Positive
4	36.8	75	98	Negative
5	38.8	112	94	Positive
6	37.2	80	98	Negative
7	39.1	114	92	Positive
8	36.7	74	99	Negative
9	38.9	111	93	Positive

Accuracy: 100.0 %

Confusion Matrix:

```
[[1 0]
 [0 2]]
```



d)

Experience	Performance Score	Training Hours	Promotion
10	95	50	Yes
8	90	45	Yes
7	88	40	Yes
3	65	20	No
2	60	18	No
9	93	48	Yes
4	70	25	No
8	89	42	Yes
2	58	15	No
6	85	38	Yes

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix
df = pd.read_excel("Dataset.xlsx",sheet_name="Dataset_4",nrows=10)
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
```

```

y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.3,random_state=42,stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(5, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=encoder.classes_,
    yticklabels=encoder.classes_
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Naive Bayes - Employee Promotion")
plt.show()
new_sample = [[9, 92, 48]]
prediction = model.predict(new_sample)
print("\nNew Sample Prediction:",encoder.inverse_transform(prediction)[0])

```

OUTPUT:

Dataset:

	Experience	Performance Score	Training Hours	Promotion
0	10	95	50	Yes
1	8	90	45	Yes
2	7	88	40	Yes
3	3	65	20	No
4	2	60	18	No
5	9	93	48	Yes
6	4	70	25	No
7	8	89	42	Yes
8	2	58	15	No
9	6	85	38	Yes

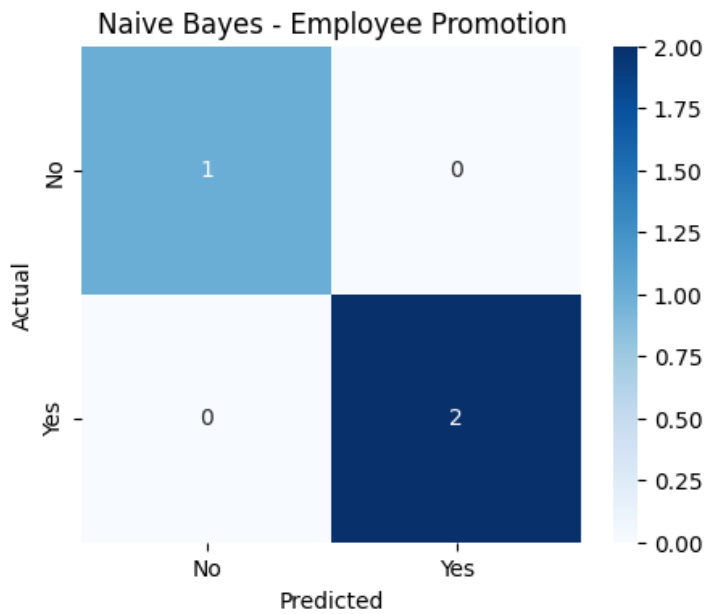
Accuracy: 100.0 %

Confusion Matrix:

```

[[1 0]
 [0 2]]

```

e)

Weight	Diameter	Sweetness	Fruit
150	7.5	90	Apple
160	7.8	88	Apple
140	7.2	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6	74	Orange
155	7.6	89	Apple
118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix

df = pd.read_excel("Dataset.xlsx",sheet_name="Dataset_5",nrows=10)
```

```
print("Dataset:")
print(df)
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test =
train_test_split(X,y,test_size=0.3,random_state=42,stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
plt.figure(figsize=(5, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=encoder.classes_,
    yticklabels=encoder.classes_
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Naive Bayes - Fruit Classification")
plt.show()
```

```

new_sample = [[155, 7.6, 89]]
prediction = model.predict(new_sample)
print("\nNew Sample Prediction:",encoder.inverse_transform(prediction)[0])

```

OUTPUT:

Dataset:

	Weight	Diameter	Sweetness	Fruit
0	150	7.5	90	Apple
1	160	7.8	88	Apple
2	140	7.2	91	Apple
3	120	5.5	70	Orange
4	125	5.8	72	Orange
5	130	6.0	74	Orange
6	155	7.6	89	Apple
7	118	5.4	69	Orange
8	148	7.4	92	Apple
9	122	5.7	71	Orange

Accuracy: 100.0 %

Confusion Matrix:

```

[[2 0]
 [0 1]]

```

